

ROCKY MOUNTAIN HIGH SPEED RAIL PROJECT MANAGEMENT PLAN

FINAL REPORT

DECEMBER 2025

GROUP 1

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Background

This report examines the Rocky Mountain Rail Authority (RMRA) High-Speed Rail Feasibility Study, a large intergovernmental transportation planning initiative in the State of Colorado. The project focuses on assessing the feasibility of implementing high-speed intercity passenger rail service along two primary corridors—the I-25 Front Range corridor (from the Wyoming border to the New Mexico border) and the I-70 corridor (from Denver International Airport to the Utah border) as well as several secondary corridors that connect key mountain communities and tourism destinations.

The study is led by the RMRA, with technical work delivered by Transportation Economics & Management Systems, Inc. (TEMS) in association with Quandel Consultants, LLC. The project adopts a Business Plan–driven feasibility approach that integrates engineering analysis, market and ridership forecasting, operations planning, and economic and financial evaluation to determine whether high-speed rail is a viable, sustainable, and fundable option for Colorado's long-term mobility needs.

Problem Statement and Project Objectives

Colorado's major north–south and east–west transportation corridors face increasing congestion, capacity constraints, and growing travel demand driven by population growth, tourism, and freight activity. Existing highway and conventional rail infrastructure are under pressure, particularly along I-25 and I-70, which serve as critical regional, interstate, and tourism arteries. Without new capacity and mode options, these corridors are expected to experience worsening travel times, reduced reliability, and negative economic and environmental impacts.

In response, the RMRA commissioned this feasibility study with the overall objective of providing a fresh, objective, and comprehensive assessment of high-speed rail as a long-term solution for these corridors. Specifically, the project aims to:

- Determine the technical and economic feasibility of implementing high-speed rail service generally within the I-25 and I-70 corridors.
- Evaluate potential technologies and service concepts that can achieve sustained speeds of 90 mph or greater, in line with Federal Railroad Administration (FRA) expectations.
- Quantify projected ridership, revenue, capital and operating costs, and financial performance, including operating ratio and cost–benefit outcomes required under FRA public–private partnership criteria.
- Identify user benefits (e.g., travel time savings and reliability), community and economic benefits (e.g., congestion relief, regional accessibility, and development opportunities), and broader societal impacts.
- Recommend preferred high-speed rail options and define the next implementation steps that RMRA and its partners should pursue, positioning Colorado to seek FRA high-speed rail corridor designation and potential federal funding.

Project Scope and Success Criteria

In-Scope Activities

The scope of this project, as analyzed in this report, is limited to activities directly related to planning and feasibility assessment, including:

- Development of a comprehensive project management plan to organize work breakdown structure, schedule, budget, roles, and communication mechanisms for the feasibility study.
- Preparation of detailed work tasks and milestones, including project management, peer review, scoping and public outreach, data collection and methodology, service scenario definition, alternatives analysis, feasibility determination, and final documentation.
- Creation of major technical databases (market, engineering, technology, and property/station databases) to support quantitative ridership, cost, and benefit modeling.
- Design and evaluation of high-speed rail alternatives, including technology options, alignments, service levels, and station strategies across primary and secondary corridors.
- Conduct of ridership, revenue, financial, and economic analyses using the COMPASS™ modeling framework and Business Plan methodology.
- Preparation of a final Feasibility and Business Plan report and associated implementation plan, summarizing findings and recommendations.

Out-of-Scope Activities

The current study does not include detailed engineering design, right-of-way acquisition, environmental impact statement (EIS) preparation, or final construction and procurement activities. It also does not commit RMRA to implementation; rather, it is intended to provide the analytical basis and strategic direction for any subsequent, more detailed phases that may be funded later.

Success Criteria

From a project management perspective, the high-speed rail feasibility study will be considered successful if it:

- Delivers a comprehensive, high-quality study on schedule and within budget, as mandated in the Project Management Plan.
- Produces results that meet FRA public–private criteria (positive operating ratio and positive cost–benefit ratio) for at least one preferred corridor option.
- Provides credible, Investment-Grade style ridership and revenue forecasts that are robust enough to support external review and potential investor/government decision-making.
- Achieves strong stakeholder and public engagement, documenting corridor needs, local expectations, and policy issues, and building regional consensus around preferred alternatives.

Assumptions and Constraints

The project is planned and executed under several critical assumptions and constraints:

- Dependence on existing studies and data – The feasibility work assumes effective coordination with prior and ongoing CDOT and I-70 Coalition studies to avoid duplication and to leverage existing technical information on alignments, station locations, and land-use considerations.

- Funding and resource limitations – The scope of analysis, especially for secondary corridors, is constrained by available budget. Screening-level methods will be used where necessary, with the understanding that more detailed work would occur only if future funding is secured.
- Time-bounded schedule and milestones – The project must adhere to a defined schedule with key deliverables (e.g., methodology report, alternatives development report, ridership and revenue report, and final report) due on specific dates; slippage in one task can impact subsequent analyses and stakeholder review windows.
- Stakeholder cooperation and right-of-way access – Successful analysis and eventual implementation assume reasonable cooperation from freight railroads, state and local agencies, and right-of-way owners, particularly where use of existing rail corridors or the CDOT highway right-of-way is under consideration.
- Managing change – The team recognizes that scope, schedule, or resource changes may arise from technical issues, stakeholder feedback, or external factors. A formal change-management procedure—including identification, documentation, impact assessment, and approval of changes—is therefore embedded within the Project Management Plan.

Project Organization & Resources

This section defines how the team is organized, who is accountable for what, how we communicate, and how people/time are allocated to deliverables. The Organization Breakdown Structure (OBS) is aligned to the WBS so each work package has a single accountable owner. The resource plan is time-phased and feeds schedule and cost baselines. Approved changes follow a lightweight decision path and are recorded in a change log.

Organization Breakdown Structure (OBS) & Roles

Org structure:

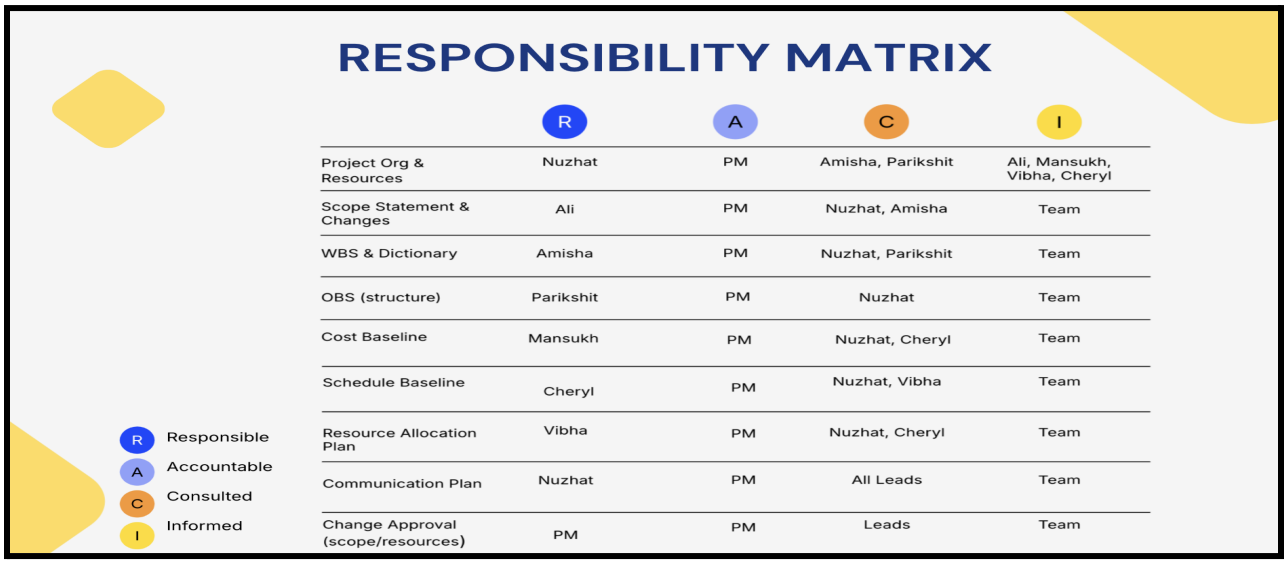
Project Manager (PM)
- Scope Lead – Ali
- Org & Resources Lead – Nuzhat
- WBS Lead – Amisha
- OBS/Control Accounts – Parikshit
- Cost/Budget – Mansukh
- Scheduling – Cheryl
- Resource Allocation – Vibha

Role summaries:

- **PM (X):** Overall delivery; owns baselines; approves changes; stakeholder reporting.
- **Ali (Scope):** Scope statement & acceptance criteria; scope change input.

- **Nuzhat (Org/Resources):** OBS↔WBS mapping, RACI, staffing plan, comms, doc control, change log.
- **Amisha (WBS):** WBS & dictionary; ensures MECE; integrates with OBS.
- **Parikshit (OBS/CA):** Maintain OBS structure; define control accounts (WBS×OBS).
- **Mansukh (Cost):** Time-phased cost baseline; track planned vs actuals.
- **Cheryl (Schedule):** Network/schedule; critical path; publishes updates.
- **Vibha (Resource Allocation):** Detailed assignments; leveling; resolves conflicts.

Responsibility Assignment (RACI):



	R	A	C	I
Project Org & Resources	Nuzhat	PM	Amisha, Parikshit	Ali, Mansukh, Vibha, Cheryl
Scope Statement & Changes	Ali	PM	Nuzhat, Amisha	Team
WBS & Dictionary	Amisha	PM	Nuzhat, Parikshit	Team
OBS (structure)	Parikshit	PM	Nuzhat	Team
Cost Baseline	Mansukh	PM	Nuzhat, Cheryl	Team
Schedule Baseline	Cheryl	PM	Nuzhat, Vibha	Team
Resource Allocation Plan	Vibha	PM	Nuzhat, Cheryl	Team
Communication Plan	Nuzhat	PM	All Leads	Team
Change Approval (scope/resources)	PM	PM	Leads	Team

Staffing & Resource Plan (time-phased)

Mapping to WBS:

Start date: 06/01/2008
Finish date: 07/14/2009
Total duration: 293 days

WBS	Task	A	R (people)	Duration (d)	Start	Finish	Effort (hrs)
1.4	Project Management Plan	Amisha	Amisha, Nuzhat	10	6/20/08	7/3/08	Amisha 40; Nuzhat 20
3.1	Scoping Workshops & Purpose/Need	Nuzhat	Nuzhat, Ali	10	8/20/08	9/2/08	Nuzhat 40; Ali 20
4.2	Market Database	Parikshit	Parikshit, Ali	15	10/15/08	11/4/08	Parikshit 90; Ali 30
6.1	Ridership Forecasts	Cherly	Cherly, Mansukh	20	1/28/09	2/24/09	Cherly 80; Mansukh 80
8.3	Draft Final Report	Vibha	Vibha, All Leads	1	7/14/09	7/14/09	Vibha 6; each lead ~3–4

Resource calendar (FTE/week):

Weeks anchored to baseline start (06/01/2008). Totals are team FTE per week; individual FTE × 40 ≈ weekly hours. Add or adjust PM (X) if you include governance effort.

Week	X (PM)	Ali	Nuzhat	Amisha	Prikshit	Mansukh	Cherly	Vibha	Total
W1 (Jun 1–7,	0.1	0.5	0.5	0.5	0.1	0	0	0	**1.7**
W2 (Jun 8–14	0.1	0.5	0.2	0.7	0.1	0	0.4	0	**2**
W3 (Jun 15–2	0.1	0.1	0.2	0.2	0.4	0.5	0.7	0.4	**2.6**
W4 (Jun 22–2	0.1	0	0.1	0	0.1	0.7	0.4	0.4	**1.8**

Communication Plan:

Information	Audience	Cadence/Timing	Method	Owner (Backup)	SLA / Where it lives	Output
Weekly status (progress/risks/next)	Team	Fri weekly (30 min)	Meeting + 1-pager	X (Nuzhat)	**Publish by EOD Fri** • /TeamDrive/Status/	Status note + action list
Issues & escalations	PM, Leads	**As needed**	Slack #apm-project + email	X (Nuzhat)	**Ack <4h, resolve/route <24h** • /Issues/Log.xlsx	Issue ticket + log update
Milestone / gate review	Instructor, Stakeholders	Per milestone	Slides + demo	**X + responsible lead**	**Deck 24h before mtg** • /Milestones/	Decision notes & sign-off
Change log update	Team	**Within 24h of approval**	Shared doc	**Nuzhat (X)**	**Log entry <24h** • /ChangeLog/ChangeLog.xlsx	Updated change record
Document releases	Team	On publish	Repo PR + tag	**Doc owner (Nuzhat)**	**Tag vX.Y** • /Docs/	Versioned artifact

Quality & Document Control:

QC matrix (Lead → Check → Approve)

Artifact	Lead	Check	Approve
PMP, Status, Minutes	Amisha	Nuzhat	X
Scoping/Outreach docs	Nuzhat	Ali	X
Methods & Existing Conditions	Amisha	Nuzhat	X
Databases (Market/Eng/Tech/Station)	Parikshit, Cherly, Ali, **Vibha (4.5)**	Amisha	X
Alternatives (5.x & 6.x)	Parikshit/Mansukh/Cherly	Amisha	X
Feasibility & Finance (7.x)	Mansukh	Amisha	X
Implementation/Business/Final (8.x)	Vibha	All Leads	X

Doc control rules: every artifact has version, date, owner, approver, change summary; obsolete versions archived; index points to latest approved.

Change & Decision Rules (execution)

Flow: Identify → Log → Impact (scope/schedule/cost/resources) → Approve (PM; escalate if major) → Implement → Communicate → Track.
Major if: milestone moves or >10% effort/cost added; otherwise minor.

On approval, update WBS/OBS/RACI + resource calendar + schedule + cost baseline within 24h.

Hand-off to Schedule & Cost

- **WBS×OBS** intersections form control accounts with a single A.
- The resource plan feeds the time-phased schedule and cost baseline.
- Once signed, this section is baselined; changes go through the log.

WBS Design

The Work Breakdown Structure is a deliverable-based, hierarchical decomposition of the total project scope. It breaks the project into progressively smaller and more manageable components so that every part of the work is clearly defined, planned, scheduled, assigned, and controlled. The WBS ensures that nothing important is overlooked and that each deliverable has an identified owner and a measurable outcome. It also serves as the foundation for developing the schedule, assigning resources, estimating costs, and later evaluating project performance.

File Automation Forms Connections — RMRA Feasibility Study – WBS & Schedule

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	Task Name	WBS Code	Assigned To	Duration	Start	Finish	Predecessors
1	Rocky Mountain Rail Feasibility Study (Master Summary)	0	All Team	293d	06/01/08	07/14/09	
2	Project Management	1	Amisha	25d	06/01/08	07/03/08	
3	Steering Committee Meetings	1.1	Amisha	5d	06/01/08	06/05/08	
4	PMC Coordination Meetings	1.2	Amisha	5d	06/06/08	06/12/08	3
5	Monthly Progress Reports	1.3	Amisha	5d	06/13/08	06/19/08	4
6	Project Management Plan	1.4	Amisha	10d	06/20/08	07/03/08	5
7	Peer Review Panels	2	Ali	33d	07/04/08	08/19/08	
8	Develop Panels	2.1	Ali	10d	07/04/08	07/17/08	6
9	First Peer Review (Data/Methods)	2.2	Ali	5d	07/18/08	07/24/08	8
10	Second Peer Review (Alternatives Analysis)	2.3	Ali	5d	07/25/08	07/31/08	9
11	Peer Review Presentations	2.4	Ali	3d	08/01/08	08/05/08	10
12	Responses to First Peer Review	2.5	Ali	5d	08/06/08	08/12/08	11
13	Responses to Second Peer Review	2.6	Ali	5d	08/13/08	08/19/08	12
14	Scoping & Public Outreach	3	Nuzhat	40d	08/20/08	10/14/08	
15	Scoping Workshops & Purpose/Need	3.1	Nuzhat	10d	08/20/08	09/02/08	13
16	Ongoing Public Input	3.2	Nuzhat	20d	09/03/08	09/30/08	15
17	Policy Outreach (Workshops)	3.3	Nuzhat	10d	10/01/08	10/14/08	16
18	Stakeholder Outreach Approach Report	3.4	Nuzhat	0	10/14/08	10/14/08	17
19	Scoping Technical Report	3.5	Nuzhat	0	10/14/08	10/14/08	18
20	Methodology & Existing Conditions	4	All Team	50d	10/14/08	12/23/08	
21	Methodology Report	4.1	Amisha	0	10/14/08	10/14/08	19
22	Market Database	4.2	Parikshit	15d	10/15/08	11/04/08	21
23	Engineering Database	4.3	Cheryl	15d	11/05/08	11/25/08	22
24	Technology Database	4.4	Ali	10d	11/26/08	12/09/08	23
25	Property/Station Database	4.5	Vibha	10d	12/10/08	12/23/08	24
26	Existing Conditions Report	4.6	Amisha	0	12/23/08	12/23/08	25
27	Preliminary Service Scenarios	5	Parikshit	25d	12/24/08	01/27/09	

	Preliminary Service Scenarios	5	Parikshit	25d	12/24/08	01/27/09	
	Identify Alternatives (routes/tech)	5.1	Parikshit	10d	12/24/08	01/06/09	26
	Develop Initial Service Concepts	5.2	Parikshit	10d	01/07/09	01/20/09	28
	Alternatives Development Workshop	5.3	Parikshit	0	01/20/09	01/20/09	29
	Alternatives Development Technical Report	5.4	Parikshit	5d	01/21/09	01/27/09	30
	Alternatives Analysis	6	All Team	50d	01/28/09	04/07/09	
	Ridership Forecasts	6.1	Cheryl	20d	01/28/09	02/24/09	31
	Revenue Forecasts	6.2	Mansukh	10d	02/25/09	03/10/09	33
	Evaluation of Alternatives	6.3	Cheryl	10d	03/11/09	03/24/09	34
	Operating & Capital Costs	6.4	Cheryl	10d	03/25/09	04/07/09	35
	Ridership & Revenue Report	6.5	Mansukh	0	04/07/09	04/07/09	36
	Alternatives Technical Report	6.6	Mansukh	0	04/07/09	04/07/09	37
	Feasibility Determination	7	All Team	50d	04/08/09	06/16/09	
	Financial Analysis (Pro forma)	7.1	Mansukh	10d	04/08/09	04/21/09	38
	User Economic Analysis	7.2	Mansukh	10d	04/22/09	05/05/09	40
	Community Economic Benefits	7.3	Mansukh	10d	05/06/09	05/19/09	41
	Financing & Funding Arrangements	7.4	Mansukh	10d	05/20/09	06/02/09	42
	Institutional Framework	7.5	Parikshit	5d	06/03/09	06/09/09	43
	Allocation of Costs & Revenues	7.6	Mansukh	5d	06/10/09	06/16/09	44
	Final Recommendation	7.7	Mansukh	0	06/16/09	06/16/09	45
	Documentation & Deliverables	8	Vibha	20d	06/17/09	07/14/09	
	Implementation Plan	8.1	Vibha	10d	06/17/09	06/30/09	46
	Business Plan	8.2	Vibha	10d	07/01/09	07/14/09	48
	Draft Final Report	8.3	Vibha	0	07/14/09	07/14/09	49
	Final Report	8.4	Vibha	0	07/14/09	07/14/09	50

Major Deliverables, Work Packages and WBS Dictionary:

The plan is organized into 5 major deliverables and 20 work packages that cover governance, scoping, baseline methods/data, alternatives development & analysis, and feasibility/business documentation. This structure aligns with our Smartsheet WBS and supports clear ownership, budgeting (control accounts), and schedule control.

Deliverable 1. Project Management & Governance

Purpose: run the project cadence, quality, and decision making.
Work packages (4):

- **1.1 Steering Committee Meetings**: governance touchpoints; decisions/minutes captured.
- **1.2 PMC Coordination Meetings**: bi-weekly internal coordination and issue tracking.
- **1.3 Monthly Progress Reports**: status, schedule/cost, risks; submitted monthly.
- **1.4 Project Management Plan (PMP)**: plan prepared and approved as project baseline.

Deliverable 2. Stakeholder Engagement & Scoping

Purpose: define purpose/need and secure stakeholder alignment.
Work packages (5):

- **3.1 Scoping Workshops & Purpose/Need**: facilitated sessions across corridors.
- **3.2 Ongoing Public Input**: continuous outreach, updates, and feedback log.

- **3.3 Policy Outreach (Workshops)**: briefings with policy/agency stakeholders.
- **3.4 Stakeholder Outreach Approach Report**: documented engagement approach.
- **3.5 Scoping Technical Report**: scoping results compiled and approved.

Deliverable 3. Baseline Data & Methodology

Purpose: establish analytical methods and current-state databases.

Work packages (5):

- **4.1 Methodology Report**: methods for demand, costs, and evaluation.
- **4.2 Market Database**: O/D, socio-economic, and network datasets.
- **4.3 Engineering Database**: ROW, geometry, and constraints.
- **4.4 Technology Database**: technology/operating speed options.
- **4.5 Property/Station Database**: stations and property value inputs.
  (Integrated into)

- **4.6 Existing Conditions Report**: synthesized baseline conditions.

Deliverable 4. Service Alternatives Development & Analysis

Purpose: Create and test candidate service concepts

Work packages (6):

- **5.1 Identify Alternatives (routes/tech)**: long-list screening to short-list.
- **5.2 Develop Initial Service Concepts**: base/improved service patterns.
- **5.3 Alternatives Development Workshop**: stakeholder/technical review session.
- **5.4 Alternatives Development Technical Report**: documented set of alternatives.
- **6.1 Ridership Forecasts**: demand modeling for short-listed options.
- **6.3 Evaluation of Alternatives**: engineering/operations feasibility assessment.
  (with supporting) **6.2 Revenue Forecasts** and **6.4 Operating & Capital Costs**; results issued as **6.5 Ridership & Revenue Report** and **6.6 Alternatives Technical Report**.

Deliverable 5. Feasibility & Business Documentation

Purpose: determine feasibility and compile implementation/business plans.

Work packages (5):

- **7.1 Financial Analysis (Pro forma)**: cash flows, NPV/IRR, funding gap.
- **7.2 User Economic Analysis**: user benefits and consumer surplus.
- **7.3 Community Economic Benefits**: jobs, income, property impacts.
- **7.5 Institutional Framework & 7.6 Allocation of Costs & Revenues**: governance and cost/revenue sharing.

- **8.1–8.4 Implementation Plan, Business Plan, Draft Final, Final Report**: packaged deliverables for approval.

Acceptance Criteria for each Deliverable:

D1. Project Management & Governance (WBS 1.x)

- **SC/PMC Meetings & Minutes** are posted within 48 hours; action items assigned with due dates and owners.
- **Monthly Progress Reports** delivered by the 5th business day; include scope/schedule/cost/risk status and agreed changes.
- **Project Management Plan (PMP)** is approved by sponsor/PM; version-controlled; becomes the baseline for scope/schedule/cost.

D2. Stakeholder Engagement & Scoping (WBS 3.x)

- **Scoping workshops** conducted per plan; attendance meets target; purpose/need statement endorsed by Steering Committee.
- **Public & policy outreach** log complete (dates, audience, issues, responses).
- **Stakeholder Outreach Approach Report** approved by sponsor.
- **Scoping Technical Report** accepted by Steering Committee; open issues list closed or assigned.

D3. Baseline Data & Methodology (WBS 4.x)

- **Methodology Report** approved (methods traceable to data/assumptions).
- **Five databases (market, engineering, technology, property/station)** delivered as cleaned, documented datasets; basic validation checks pass (completeness, formats, referential integrity).
- **Existing Conditions Report** integrates 4.1– 4.5 and is approved for use in alternatives development.

D4. Service Alternatives Development & Analysis (WBS 5.x–6.x)

- **Alternatives shortlist & service concepts** documented and approved after workshop.
- **Models run**: ridership, revenue, operating & capital cost results reproducible; assumptions logged.
- **Evaluation matrix** completed against agreed criteria; recommendations traceable to evidence.
- **Ridership & Revenue Report** and **Alternatives Technical Report** approved; peer-review comments resolved or dispositioned.

D5. Feasibility & Business Documentation (WBS 7.x–8.x)

- **Financial pro forma (NPV/IRR)** consistent with 6.x results; review checklist passed.
- **User & community economic analyses** completed with stated assumptions; QA checks signed.
- **Institutional framework & cost/revenue allocation** options evaluated; preferred approach endorsed.
- **Implementation Plan / Business Plan / Final Report** issued; all prior comments closed; sponsor sign-off recorded.

4. OBS Design and Control Account Matrix

Organizational Breakdown Structure (OBS)

The Organizational Breakdown Structure (OBS) defines how our project team is organized to deliver the scope identified in the WBS. It clarifies reporting relationships, functional responsibilities, and the structure through which work is coordinated and controlled.

For this project, we used a three-level OBS hierarchy:

Level 1 – Project Sponsor

- **Dr. Steve Peng** serves as the overall project sponsor and key stakeholder.
  He provides strategic guidance, defines expectations, and approves major scope decisions.

Level 2 – Project Manager

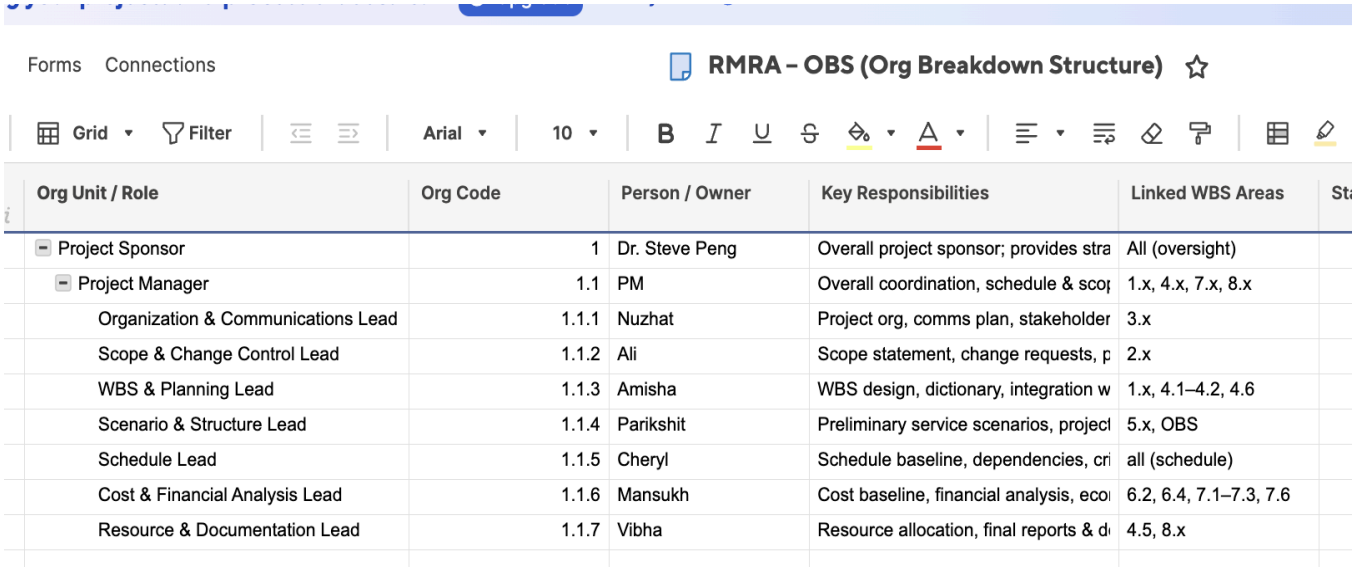
- **PM** serves as the Project Manager, responsible for overall coordination, integration of deliverables, managing scope and schedule, and ensuring the team stays aligned with project objectives and stakeholder expectations.

Level 3 – Functional Leads

Reporting to the Project Manager are seven functional leads whose responsibilities align directly with major WBS deliverable groups:

- **Nuzhat – Organization & Communications Lead**
  Responsible for stakeholder communication, public outreach, and project organization tasks.
- **Ali – Scope & Change Control Lead**
  Manages scope development, peer review processes, and change requests as they arise.
- **Parikshit – Scenario & Structure Lead**
  Oversees preliminary service scenarios, early design structure, and helps refine the OBS itself.

- **Amisha – WBS & Planning Lead**
  Designs and maintains the WBS, WBS dictionary, and integrates it with planning activities.
- **Cheryl – Schedule Lead**
  Owns the project schedule, dependencies, critical path monitoring, and schedule updates.
- **Mansukh – Cost & Financial Analysis Lead**
  Develops the cost baseline and performs financial and economic analyses.
- **Vibha – Resource & Documentation Lead**
  Responsible for resource allocation tracking and production of final project deliverables.



Forms Connections

RMRA – OBS (Org Breakdown Structure)

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Org Unit / Role	Org Code	Person / Owner	Key Responsibilities	Linked WBS Areas	St
Project Sponsor	1	Dr. Steve Peng	Overall project sponsor; provides stra	All (oversight)	
Project Manager	1.1	PM	Overall coordination, schedule & scop	1.x, 4.x, 7.x, 8.x	
Organization & Communications Lead	1.1.1	Nuzhat	Project org, comms plan, stakeholder	3.x	
Scope & Change Control Lead	1.1.2	Ali	Scope statement, change requests, p	2.x	
WBS & Planning Lead	1.1.3	Amisha	WBS design, dictionary, integration w	1.x, 4.1–4.2, 4.6	
Scenario & Structure Lead	1.1.4	Parikshit	Preliminary service scenarios, project	5.x, OBS	
Schedule Lead	1.1.5	Cheryl	Schedule baseline, dependencies, cri	all (schedule)	
Cost & Financial Analysis Lead	1.1.6	Mansukh	Cost baseline, financial analysis, eco	6.2, 6.4, 7.1–7.3, 7.6	
Resource & Documentation Lead	1.1.7	Vibha	Resource allocation, final reports & d	4.5, 8.x	

WBS/OBS Integration Matrix (Control Accounts)

To link the organizational structure with the project scope, we developed a WBS/OBS Integration Matrix, also known as the Control Account Matrix. This matrix identifies Control Accounts—management control points where scope, schedule, and cost responsibility intersect.

Purpose of the Control Account Matrix

- Ensures every major WBS work package has a single accountable owner.
- Provides a framework for cost tracking, variance analysis, and earned value concepts.
- Aligns each deliverable to the correct functional unit within the OBS.
- Supports transparent performance management and smoother integration during project execution.

Control Account Matrix

CA ID	WBS Code	Work Package	OBS Role	Person Responsible
CA-01	1.1	Steering Committee Meetings	Project Manager	**Amisha**
CA-02	1.3	Monthly Progress Reports	Project Manager	**Amisha**
CA-03	2.1	Develop Panels	Scope & Change Lead	**Ali**
CA-04	2.2	First Peer Review	Scope & Change Lead	**Ali**
CA-05	2.3	Second Peer Review	Scope & Change Lead	**Ali**
CA-06	3.1	Scoping Workshops & Purpose/Need	Org & Comms Lead	**Nuzhat**
CA-07	3.2	Public Input	Org & Comms Lead	**Nuzhat**
CA-08	4.1	Methodology Report	WBS & Planning Lead	**Amisha**
CA-09	4.2	Market Database	WBS & Planning Lead	**Amisha**
CA-10	4.3	Engineering Database	Schedule Lead	**Cheryl**
CA-11	4.4	Technology Database	Scope & Change Lead	**Ali**
CA-12	4.5	Property/Station Database	Documentation Lead	**Vibha**
CA-13	4.6	Existing Conditions Report	WBS & Planning Lead	**Amisha**
CA-14	5.1	Identify Alternatives	Scenario Lead	**Parikshit**
CA-15	5.3	Alternatives Workshop	Scenario Lead	**Parikshit**
CA-16	6.1	Ridership Forecasts	Schedule Lead	**Cheryl**
CA-17	6.4	Operating & Capital Costs	Cost Lead	**Mansukh**
CA-18	7.1	Financial Analysis	Cost Lead	**Mansukh**
CA-19	7.5	Institutional Framework	Scenario Lead	**Parikshit**
CA-20	8.4	Final Report	Documentation Lead	**Vibha**

Cost Analysis and Budgeting

Estimating method used

- Bottom-up labor for detailed tasks: for each WBS work package, estimate hours by role (from your Staffing & Resource Plan).
- Analogous / apportion for small or similar tasks: scale from a known task or allocate proportionally across a deliverable.
- Indirect (overhead/G&A) applied as a % of direct labor.
- Contingency applied to (Direct + Indirect) based on uncertainty of the task/deliverable.

Formula stack:

Direct labor ($) = Σ (hours_role × rate_role)
Indirect ($) = Direct × Overhead%
Contingency ($) = (Direct + Indirect) × Contingency%
Total ($) = Direct + Indirect + Contingency

Default inputs:

Hourly rates: X=$90, Ali=$80, Nuzhat=$80, Amisha=$85, Parikshit=$85, Mansukh=$90, Cherly=$85, Vibha=$80/hour.
Overhead (indirects)=30%, Contingency=15%.

Budget by Control Account

Deliverable (Control Account)	Duration (d)	Direct $	Indirect (30%)	Contingency (15% of D+I)	Total $
D1 Project Mgmt & Governance	25	33,957	10,187	6,622	50,766
D2 Peer Review Panels	33	44,827	13,448	8,741	67,016
D3 Scoping & Outreach	40	54,335	16,301	10,595	81,231
D4 Methodology & Existing Conditions	50	67,919	20,376	13,244	101,539
D5 Preliminary Service Scenarios	25	33,957	10,187	6,622	50,766

D6 Alternatives Analysis	50	67,919	20,376	13,244	101,539
D7 Feasibility Determination	50	67,919	20,376	13,244	101,539
D8 Documentation & Deliverables	20	27,167	8,150	5,298	40,615
Totals (BAC)	293	398,000	119,400	77,610	$595,010

Budget at Completion (BAC): $595,010.

(BAC = Direct + Indirect + Contingency across all control accounts.)

D1 Project Management & Governance (Direct $ = $33,957, 25d)

WBS share: 1.1=5d (20%), 1.2=5d (20%), 1.3=5d (20%), 1.4=10d (40%).

WBS	Task	Share	Direct $	Indirect 30%	Cont. 15%	Total $
1.1	Steering Committee Mtgs	20%	6,791	2,037	1,320	10,148
1.2	PMC Coordination Mtgs	20%	6,791	2,037	1,320	10,148
1.3	Monthly Progress Reports	20%	6,791	2,037	1,320	10,148
1.4	Project Management Plan	40%	13,583	4,075	2,641	20,322
D1 total		100%	33.957	10,187	6,622	50,766

D2 — Peer Review Panels (Direct $ = $44,827; Total = $67,016)

WBS	Task	Share of	Direct $	Indirect	Cont. (15% of	Total

		D2 (by Duration)		30%	D+I)	$
2.1	Develop Panels	10/33 ≈ 30%	13,584	4,075	2,651	20,310
2.2	First Peer Review (Data/Methods)	5/33 ≈ 15%	6,792	2,038	1,324	10,154
2.3	Second Peer Review (Alternatives Analysis)	5/33 ≈ 15%	6,792	2,038	1,324	10,154
2.4	Peer Review Presentations	3/33 ≈ 9%	4,075	1,223	795	6,092
2.5	Responses to First Peer Review	5/33 ≈ 15%	6,792	2,038	1,324	10,154
2.6	Responses to Second Peer Review	5/33 ≈ 15%	6,792	2,038	1,324	10,154
D2 total			44,827	13,448	8,741	67,016

D3 — Scoping & Outreach (Direct $ = 54,335; Total = $81,231)

WBS	Task	Share of D3 (by Duration)	Direct $	Indirect 30%	Cont. (15% of D+I)	Total $
3.1	Scoping Workshops & Purpose/Need	25%	13,584	4,075	2,649	20,308
3.2	Ongoing Public Input	50%	27,167	8,150	5,298	40,615
3.3	Policy Outreach (Workshops)	25%	13,584	4,075	2,649	20,308
3.4 - 3.5	Reports (0d)	0%	~0	~0	~0	~0

D3 total		100%	54,335	16,301	10,595	81,231

D4 — Methodology & Existing Conditions (Direct $ = 67,919; Total = $101,539)

WBS	Task	Share of D4 (by Duration)	Direct $	Indirect 30%	Cont. (15% of D+I)	Total $
4.2	Market Database	**30%**	**20,376**	**6,113**	**3,973**	**30,462**
4.3	Engineering Database	**30%**	**20,376**	**6,113**	**3,973**	**30,462**
4.4	Technology Database	**20%**	**13,584**	**4,075**	**2,649**	**20,308**
4.5	Property/Station Database	**20%**	**13,583**	**4,075**	**2,649**	**20,307**
4.1 & 4.6	Methodology/Existing Cond. Reports (0d)	**0%**	**~0**	**~0**	**~0**	**~0**
D4 total		**100%**	**67,919**	**20,376**	**13,244**	**101,539**

D5 — Preliminary Service Scenarios (Direct $ = 33,957; Total = $50,766)

WBS	Task	Share of D5 (by Duration)	Direct $	Indirect 30%	Cont. (15% of D+I)	Total $
5.1	Identify Alternatives (routes/tech)	40%	13,583	4,075	2,649	20,307
5.2	Develop Initial Service Concepts	40%	13,582	4,075	2,649	20,307
5.3	Alternatives Development Technical Report	20%	6,791	2,037	1,324	10,152
D5		100%	33,957	10,187	6,622	50,766

total						

D6 Alternatives Analysis (total Direct $ = $67,919, 50d)

WBS share: 6.1=20d (40%), 6.2=10d (20%), 6.3=10d (20%), 6.4=10d (20%), 6.5=0d, 6.6=0d.

WBS	Task	Share	Direct $	Indirect 30%	Cont. 15%	Total $
6.1	Ridership Forecasts	40%	27,168	8,150	5,281	40,599
6.2	Revenue Forecasts	20%	13,584	4,075	2,640	20,299
6.3	Evaluation of Alternatives	20%	13,584	4,075	2,640	20,299
6.4	Operating & Capital Costs	20%	13,584	4,075	2,640	20,299
6.5–6.6	Reports (0d)	0%	~0	~0	~0	~0
D6 total		100%	67,919	20,376	13,244	101,539

D7 — Feasibility Determination (Direct $ = $67,919; Total = $101,539)

WBS	Task	Share of D7	Direct $	Indirect (30%)	Cont. (15% of D+I)	Total $
7.1	Financial Analysis (Pro forma)	10/50 = 20%	13,584	4,075	2,649	20,308
7.2	User Economic Analysis	10/50 = 20%	13,584	4,075	2,649	20,308
7.3	Community Economic Benefits	10/50 = 20%	13,584	4,075	2,649	20,308
7.4	Financing & Funding Arrangements	10/50 = 20%	13,584	4,075	2,649	20,308
7.5	Institutional Framework	5/50 = 10%	6,792	2,038	1,324	10,154

7.6	Allocation of Costs & Revenues	5/50 = 10%	6,792	2,038	1,324	10,154
7.7	Final Recommendation	0%	~0	~0	~0	~0
D7 total			67,919	20,376	13,244	101,539

D8 — Documentation & Deliverables (Direct $ = 27,167; Total = $40,615)

WBS	Task	Share of D8	Direct $	Indirect 30%	Cont. (15% of D+I)	Total $
8.1	Implementation Plan	50%	13,584	4,075	2,649	20,308
8.2	Business Plan	50%	13,583	4,075	2,649	20,307
8.3-8.4	Draft/Final Report (0d)	0%	~0	~0	~0	~0
D8 total		100%	27,167	8,150	5,298	40,615

Direct / Indirect / Contingency policy

- Direct: labor only (include small travel/printing if needed within each work package).
- Indirect: uniform 30% on direct.
- Contingency: 15% on *(Direct+Indirect)*

We estimated costs using bottom-up hours where known and apportion/analogous estimating against WBS durations for the rest. We applied 30% indirects and 15% contingency to direct labor to form the cost baseline. Budgets roll up by work package to Control Accounts (deliverables) and to project BAC ($595,010), aligning cost with the schedule and resource plan.

Scheduling & Resource Allocation

Effective scheduling and resource allocation ensure that the Rocky Mountain Rail Feasibility Study can be completed within the 293-day baseline (June 1, 2008 – July 14, 2009). This section summarizes how the Work Breakdown Structure (WBS) was translated into the activity schedule,

how dependencies were mapped through a network diagram, how the critical path was identified, and how resources were assigned and leveled to keep the schedule feasible.

6.1 Activity List, Durations, and Logical Dependencies

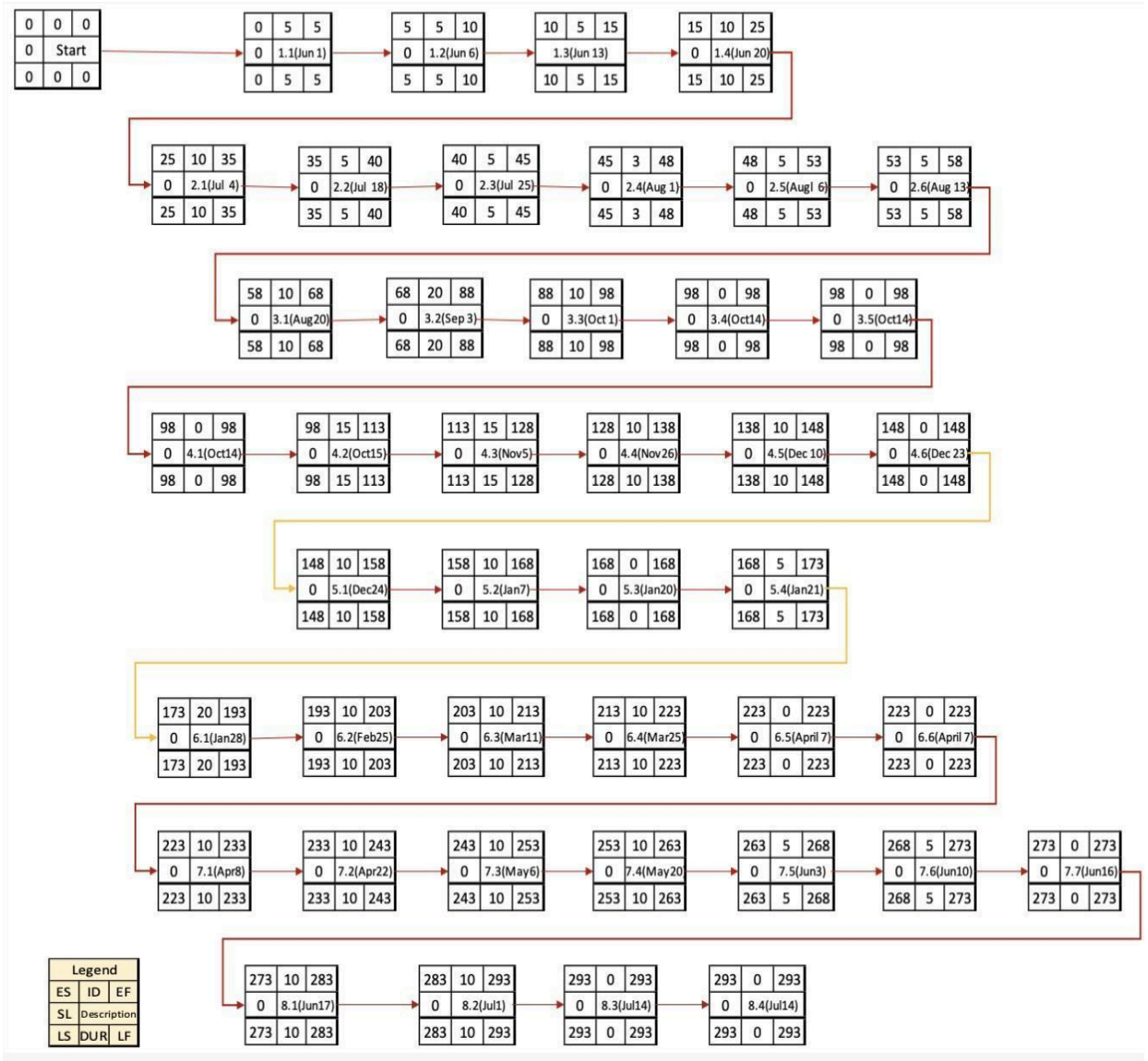
All major work packages in the WBS were decomposed into discrete schedule activities. Durations, start/finish dates, and predecessors were extracted from the Smartsheet task dataset (baseline: 06/01/08–07/14/09). Activities were grouped by WBS to support schedule rollups. A condensed summary of the major activity groups is shown below:

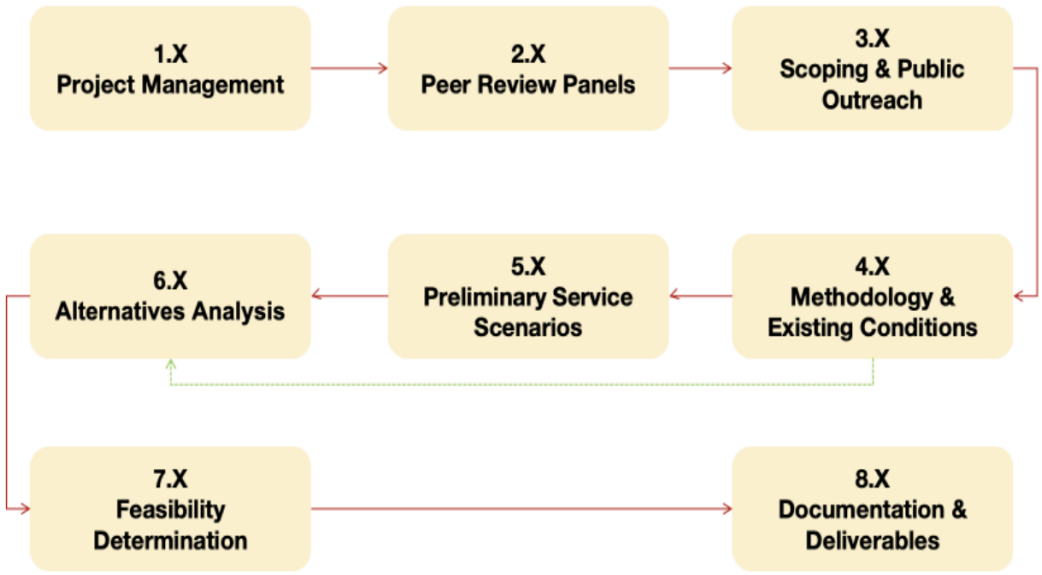
WBS	Activity Name	Duration	Predecessor	Owner
1.1–1.4	Project Management Activities	25 days	Sequential (1→2→3→4)	Ali/Amisha
2.1–2.4	Peer Review Process	33 days	1.4	Ali
3.x	Scoping & Public Outreach	104 days	2.x	Nuzhat
4.x	Methodology + Databases	156 days	3.5	Amisha/Cheryl/Vibha
5.x–6.x	Alternatives + Ridership/Revenue	106 days	4.6	Parikshit/Cheryl
7.x–8.x	Feasibility + Final Reports	84 days	6.x	Mansukh/Vibha

These data form the basis of the schedule network and were used to calculate Early Start (ES), Early Finish (EF), Late Start (LS), Late Finish (LF), and total float for each activity.

6.2 Network Diagram and Critical Path Analysis

The full network diagram contains more than 70 linked activities and reflects the complete logical sequencing of the project. Mandatory dependencies create a single dominant path through the analysis, beginning with project management tasks and continuing through peer review, scoping, methodology, alternatives development, ridership and revenue analysis, feasibility evaluations, and reporting.


0 0 0
0 Start 0
0 0 0
0 5 5
0 1.1(Jun 1) 0
0 5 5
5 5 10
0 1.2(Jun 6) 0
5 5 10
10 5 15
1.3(Jun 13)
10 5 15
15 10 25
0 1.4(Jun 20) 0
15 10 25
25 10 35
0 2.1(Jul 4) 0
25 10 35
35 5 40
0 2.2(Jul 18) 0
35 5 40
40 5 45
0 2.3(Jul 25) 0
40 5 45
45 3 48
0 2.4(Aug 1) 0
45 3 48
48 5 53
0 2.5(Augl 6) 0
48 5 53
53 5 58
0 2.6(Aug 13) 0
53 5 58
58 10 68
0 3.1(Aug20) 0
58 10 68
68 20 88
0 3.2(Sep 3) 0
68 20 88
88 10 98
0 3.3(Oct 1) 0
88 10 98
98 0 98
0 3.4(Oct14) 0
98 0 98
98 0 98
0 3.5(Oct14) 0
98 0 98
98 0 98
0 4.1(Oct14) 0
98 0 98
98 15 113
0 4.2(Oct15) 0
98 15 113
113 15 128
0 4.3(Nov5) 0
113 15 128
128 10 138
0 4.4(Nov26) 0
128 10 138
138 10 148
0 4.5(Dec 10) 0
138 10 148
148 0 148
0 4.6(Dec 23) 0
148 0 148
148 10 158
0 5.1(Dec24) 0
148 10 158
158 10 168
0 5.2(Jan7) 0
158 10 168
168 0 168
0 5.3(Jan20) 0
168 0 168
168 5 173
0 5.4(Jan21) 0
168 5 173
173 20 193
0 6.1(Jan28) 0
173 20 193
193 10 203
0 6.2(Feb25) 0
193 10 203
203 10 213
0 6.3(Mar11) 0
203 10 213
213 10 223
0 6.4(Mar25) 0
213 10 223
223 0 223
0 6.5(April 7) 0
223 0 223
223 0 223
0 6.6(April 7) 0
223 0 223
223 10 233
0 7.1(Apr8) 0
223 10 233
233 10 243
0 7.2(Apr22) 0
233 10 243
243 10 253
0 7.3(May6) 0
243 10 253
253 10 263
0 7.4(May20 0
253 10 263
263 5 268
0 7.5(Jun3) 0
263 5 268
268 5 273
0 7.6(Jun10) 0
268 5 273
273 0 273
0 7.7(Jun16) 0
273 0 273
273 10 283
0 8.1(Jun17) 0
273 10 283
283 10 293
0 8.2(Jul1) 0
283 10 293
293 0 293
0 8.3(Jul14) 0
293 0 293
293 0 293
0 8.4(Jul14) 0
293 0 293
Legend
ES ID EF
SL Description
LS DUR LF



Critical Path Summary

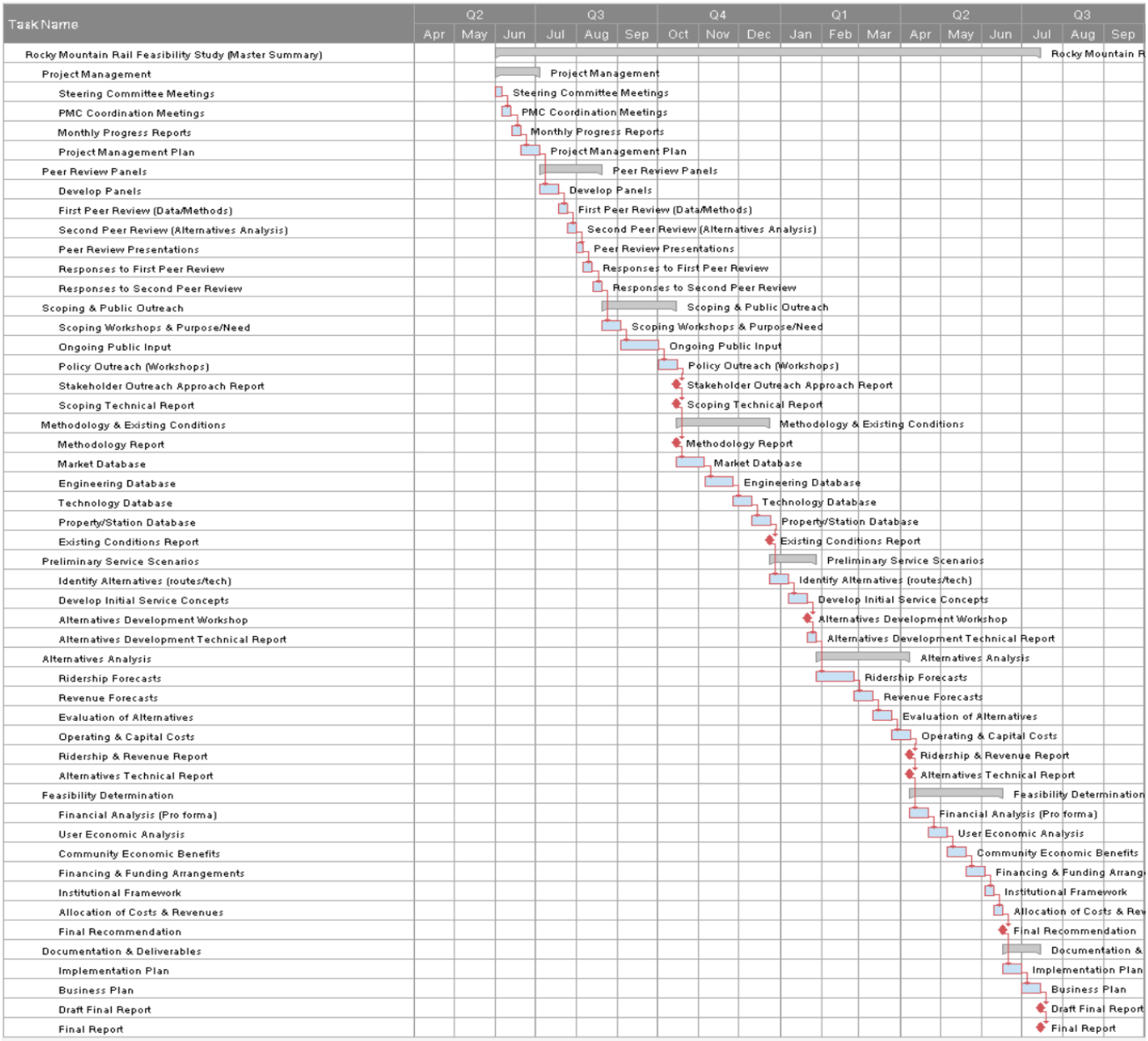
Analysis of ES/EF/LS/LF values shows that the critical path spans WBS Groups 1.x → 2.x → 3.x → 4.x → 5.x → 6.x → 7.x → 8.x, with zero float across these deliverables.

Key characteristics:

- WBS 4.x (Methodology + Databases) dominates total duration and is the longest technical block in the schedule.
- 6.x Alternatives Analysis cannot begin until all five baseline databases are complete.
- Any delay in 4.x or 6.x automatically cascades into feasibility milestones and final documentation.
- Noncritical outreach tasks (e.g., 3.2 Ongoing Public Input and 3.3 Policy Workshops) contain float and were leveraged for leveling.
- The critical path determines the minimum project duration of 293 days.

6.3 Gantt Chart and Resource Assignment

A Gantt chart was produced to illustrate schedule flow, activity overlaps, and critical-path dependencies.



Key observations:

- Activities in WBS 4.x and 6.x drive the most significant workload concentration.
- Outreach tasks, documentation steps, and reports in 7.x–8.x overlap but remain mostly noncritical.
- Each task is assigned a single accountable owner per the WBS/OBS integration (Control Accounts), consistent with the Responsibility Matrix shown earlier (pp. 4–5).
- Team FTE loading aligns with the time-phased resource calendar presented in the Resource Plan (pp. 5–6), preventing overextension during high-demand periods.

Resource Allocation

Resource allocation is the process of assigning and managing people, budget, and time needed to complete project work. In line with PMBOK, this involves determining what resources are required, in what quantities, and when across the project lifecycle (PMI, 2021). For the RMRA High-Speed Rail Feasibility Study, resource allocation focuses on strategically distributing human, time, and financial resources to support all major deliverables and maintain the schedule and cost baselines.

Project Resource Summary

- Total project budget: $1,249,763
- Labor budget: $1,100,663 (88.1%)
- Expense budget: $149,100 (11.9%)
- Team size: 8 members (including Project Manager)
- Project duration: 293 days (June 1, 2008 – July 14, 2009)

The high share of labor cost reflects the knowledge-intensive nature of a feasibility study, relying on technical, economic, and planning expertise. Expenses primarily cover travel, workshops, and coordination with agencies such as CDOT and the I-70 Coalition.

Resource Leveling and Constraints

The RMRA project employs a four-part resource leveling strategy to maintain balance throughout the project lifecycle:

Workload Balancing- Weekly Full-Time Equivalent (FTE) allocations are monitored for each team member. When any individual exceeds 1.0 FTE capacity, tasks are redistributed to other team members with available capacity. This approach prevents burnout and maintains work quality by ensuring no single resource is overextended.

Critical Path Priority- Activities on the project's critical path receive first claim on additional resources. This strategy protects the schedule baseline by ensuring that tasks with zero float receive adequate attention. Critical path activities include PMP approval, peer reviews, and final report delivery milestones**.**

Cross-Training -Team members are trained on adjacent tasks to enable flexible reallocation when conflicts arise. For example, Amisha can support data collection tasks if Parikshit is overloaded, and Mansukh can assist with alternatives analysis during peak periods. This redundancy provides flexibility without requiring external resources.

Buffer Management- A 15% contingency reserve of hours is maintained for unplanned work and scope changes. This buffer was instrumental in absorbing the additional workday required for the DIA station option change without impacting the project timeline or requiring a budget increase.

Identified Resource Conflicts

The RMRA project identified three significant resource conflicts during the planning phase:

Conflict	Period	Issue	Impact
Peak Demand Overlap	January 2009	Ridership analysis and Peer Review #2 scheduled concurrently	High
Workshop Scheduling	October 2008	Multiple team members required for alternatives workshop	Medium
Documentation Crunch	April-May 2009	Multiple reports (Business Plan, Implementation Plan, Final Report) due simultaneously	High

Conflict Resolution Strategies

Each identified conflict requires a tailored resolution strategy. The following approaches were developed to address the specific constraints of each conflict:

Peak Demand Overlap Resolution (January 2009)

The concurrent scheduling of ridership/revenue analysis and Peer Review #2 created the highest resource demand of the project. The resolution strategy involves staggering deliverables and assigning specialized focus areas to team members. Amisha will concentrate exclusively on the ridership analysis while other team members handle peer review preparation and coordination. This division of labor ensures that both critical activities receive adequate attention without requiring additional resources.

Workshop Scheduling Resolution (October 2008)

The alternatives workshop requires the simultaneous participation of multiple team members, creating a temporary spike in resource demand. The resolution involves treating the workshop as an all-hands event with adjusted workloads before and after the workshop period. Team members complete advance work in the weeks prior and defer non-critical tasks until after the workshop. This approach smooths the resource demand curve while ensuring full participation.

Documentation Crunch Resolution (April-May 2009)

The final phase requires simultaneous production of multiple major deliverables. The resolution strategy employs parallel workstreams with the Project Manager providing prioritization guidance. Each deliverable is assigned a lead author with supporting contributors. If necessary, daily working hours may be extended during this period. The 15% buffer reserve provides additional capacity if the crunch exceeds planned levels.

Risk Register and Assessment

Risk Identification Process

Risk identification was conducted through multiple techniques to ensure comprehensive coverage:

- Brainstorming sessions with the full project team
- Expert interviews with transportation planning specialists
- Review of historical data from similar feasibility studies
- Stakeholder consultation to identify external risks
- Analysis of project assumptions and constraints

The identification process categorized risks into six categories: Stakeholder, Schedule, Cost, Technical, Resource, External, and Quality. This categorization facilitates analysis and ensures that risks across all project dimensions are considered.

Risk Register

The risk register is the primary tool for documenting identified risks and tracking their status throughout the project. Each risk is assigned a unique identifier (R01, R02, etc.) for tracking purposes. The register includes the risk description, category, probability and impact ratings, calculated risk score, and assigned owner.

Risk ID	Risk Description	Category	Prob.	Impact	Score	Owner
R01	Stakeholder resistance to proposed rail alignments	Stakeholder	4	4	16	Ali
R02	Delays in data collection from partner agencies	Schedule	3	4	12	Amisha
R03	Budget overrun due to expanded scope	Cost	3	4	12	Mansukh
R04	Technical complexity in I-70 mountain corridor	Technical	4	3	12	Amisha
R05	Key team member unavailability	Resource	3	3	9	Nuzhat
R06	Peer review panel delays or conflicts	Schedule	3	3	9	PM
R07	Ridership model assumptions challenged by FRA	Technical	3	4	12	Amisha
R08	Insufficient public participation in workshops	Stakeholder	3	3	9	Ali

R09	Right-of-way data incomplete or inaccurate	Technical	3	3	9	Parikshit
R10	Changes in federal/state funding priorities	External	2	4	8	PM
R11	Coordination conflicts with parallel studies	Schedule	3	3	9	Nuzhat
R12	Quality issues in deliverable documentation	Quality	2	3	6	Cheryl

Probability × Impact Matrix

The Probability × Impact (P×I) matrix is a tool for prioritizing risks based on their likelihood of occurrence and potential effect on project objectives. This approach aligns with PMBOK's recommendation for qualitative risk analysis (PMI, 2021).

Risk Score Calculation

The risk score is calculated by multiplying the probability rating by the impact rating:

Risk Score = Probability × Impact

Scores range from 1 (Very Low probability × Very Low impact) to 25 (Very High probability × Very High impact). For example, R01 (Stakeholder resistance) has a probability of 4 (High) and impact of 4 (High), resulting in a risk score of 16.

Risk Classification and Prioritization

Risks are classified into four zones based on their calculated scores:

Zone	Score Range	Required Action
Critical (Red)	15-25	Immediate action required; escalate to Project Manager and Steering Committee
High (Orange)	10-14	Active management required; develop and implement mitigation plans
Medium (Yellow)	5-9	Monitor closely; prepare contingency plans
Low (Green)	1-4	Accept and monitor; minimal intervention required

Risk Distribution Summary

The RMRA project risk profile is distributed as follows:

- Critical (15-25): 1 risk (R01: Stakeholder resistance)

- High (10-14): 4 risks (R02, R03, R04, R07)
- Medium (5-9): 6 risks (R05, R06, R08, R09, R10, R11)
- Low (1-4): 1 risk (R12)

The presence of one critical risk (R01: Stakeholder resistance) indicates that stakeholder management is a primary concern for this project. The cluster of four high-priority risks related to data collection, budget, technical complexity, and FRA requirements reflects the challenging nature of the feasibility study.

Risk Response Strategy

Strategy Distribution
The RMRA project employs the following distribution of response strategies:

Strategy	Count	Applicable Risks
Mitigate	10	R01, R02, R03, R05, R06, R07, R08, R09, R11, R12
Accept + Mitigate	1	R04
Transfer	1	R10
Avoid	0	None

Mitigation is the predominant strategy (10 of 12 risks) because most identified risks can be addressed through proactive management actions. The Accept + Mitigate approach for R04 (Technical complexity) acknowledges that the I-70 mountain corridor presents inherent challenges that cannot be eliminated but can be managed. Transfer is used for R10 (Federal funding changes) because this external risk is beyond the project's direct control.

Risk Response Matrix – High-Priority Risks (Summary)

The risk response matrix defines strategies, key actions, contingency plans, and triggers for each major risk:

- **R01 – Stakeholder Resistance (Score 16, Critical)**
  Strategy: Mitigate
  Key Actions: Early engagement through scoping workshops, Regular communication updates to all stakeholder groups, Explicitly address community concerns in alternatives analysis, Document and respond to all feedback received.
  Contingency: Escalate to the Steering Committee if >2 stakeholder groups file formal objections.
- **R02 – Data Collection Delays (Score 12, High)**
  Strategy: Mitigate
  Key Actions: Establish data sharing agreements upfront with partner agencies, Run parallel data collection streams where possible, Hold weekly coordination calls with

CDOT and I-70 Coalition, Identify alternative data sources as backup.
Contingency: Use proxy data or request schedule adjustment from PMC if data are >2 weeks late.

- **R03 – Budget Overrun (Score 12, High)**
  Strategy: **Mitigate**
  Key Actions: Monthly budget tracking against baseline, Early warning trigger at 80% spend threshold, All scope additions must go through change control process, Regular cost forecasting and variance analysis.
  Contingency: Request contingency funds and/or descope non-critical work if actual spend exceeds 90% of task budget.

- **R04 – Technical Complexity (Score 12, High)**
  Strategy: **Accept + Mitigate**
  Key Actions: Engage tunnel engineering experts early in the project, Validate approach through peer review panels, Use phased analysis approach to prevent blocking, Document assumptions and limitations clearly.
  Contingency: Extend timelines for specific technical tasks if effort exceeds estimates by 50%.

- **R07 – Ridership Assumptions (Score 12, High)**
  Strategy: **Mitigate**
  Key Actions: Strictly use FRA-approved methodologies, Document all assumptions clearly and transparently, Run sensitivity analysis on key variables, Present range of scenarios rather than single point estimates.
  Contingency: Provide revised scenario ranges and seek FRA pre-review if initial FRA feedback is negative.

Risk Monitoring and Control

Risk monitoring and control is an ongoing process throughout the project lifecycle. The RMRA project implements the following monitoring structure:

Review Type	Frequency	Participants	Focus
Risk Status Update	Weekly	PM, Risk Owner	Status of active risks, new risk identification
Risk Register Review	Monthly	Full Team	Update probability/impact, close resolved risks
Risk Report	Monthly	PM to Steering Committee	High/critical risks, mitigation status, trends

Risk Metrics

The following metrics are tracked to assess risk management effectiveness:

Metric	Target	Current Status
Critical risks (score 15+)	0	1 (R01)
High risks (score 10-14)	≤ 3	4

Risks with mitigation plans	100%	100%
Risk review compliance	100%	On Track

Escalation Procedures
A clear escalation path ensures that risks are addressed at the appropriate management level based on their severity:

Risk Level	Score Range	Escalation To
Low	1-4	Risk Owner manages independently
Medium	5-9	Project Manager oversight
High	10-14	Project Management Consultant (PMC)
Critical	15-25	Steering Committee / RMRA Board

The escalation process ensures that decision-making authority matches risk severity. Critical risks that threaten project success are escalated to the RMRA Board, while low-severity risks are managed at the operational level by assigned risk owners.

PART 2 – PROJECT EXECUTION

- Underground Station

1. Description of Change (What Happened)

At the start of our project, our team planned to include a standard, above-ground station at Denver International Airport as part of the route. It fit well with our original assumptions, our engineering model, and the overall scope of the feasibility study.

However, during one of our stakeholder review meetings, our stakeholder Dr. Steve Peng suggested that instead of a regular surface station, we should explore building an underground DIA station. He emphasized that an underground design could support long-term airport expansion and offer a more realistic representation of future transportation needs.

This request immediately shifted the direction of our work. An underground station introduces very different engineering, safety, and passenger-flow requirements compared to an

above-ground design. It also affects how the rail alignment connects to the airport and how we assess environmental and construction impacts.

Because this change was not part of our original scope, we had to revisit several components of our project: the station layout, the engineering assumptions, the constraints database, and even our outreach plan, since new airport teams and technical experts now needed to be involved.

Overall, this unexpected change added complexity but also increased the value of the study, allowing us to produce an option that aligns more closely with long-term infrastructure planning and of course, it made our stakeholder, Dr. Peng, very happy.

2. Approaches for Handling the Change (How We Managed It)

Once Dr. Steve Peng requested that the DIA station be redesigned as an underground station, our team treated it as a formal project change rather than an informal suggestion. Even though the request came during a discussion, we followed a structured change-management approach to make sure the shift didn't disrupt our overall timeline or create confusion.

First, we documented the change clearly so the entire team understood what was being asked. Then we assessed the impact by reviewing how an underground station would affect engineering assumptions, passenger-flow design, stakeholder involvement, and future tasks in our schedule. Since the change expanded scope but had a manageable effect on resources, we agreed it was feasible to move forward.

Next, we updated the relevant work packages, especially those connected to station design, engineering databases, and alternatives development. This meant adjusting the WBS, updating dependencies, and making sure our schedule still flowed smoothly. We also coordinated with new stakeholders such as airport facilities teams and infrastructure specialists because an underground design introduces additional considerations like tunneling, ventilation, and safety standards.

Communication was also a key part of our approach. We shared the change with our internal team, aligned expectations in our progress discussions, and noted the update in our project documentation. By keeping everyone informed, the team stayed coordinated and no one was surprised by shifts in workload or responsibilities.

Throughout the process, we continued to monitor the impact of the change to ensure it stayed within our baseline tolerances. Overall, we handled the shift systematically: clarify, evaluate, approve, implement, and communicate which helped us integrate the new underground-station requirement smoothly without losing momentum on the project.

3. Risk Monitoring and Control (Updated Risks After the Change)

When the underground station requirement was introduced, one of the first things we did was revisit our risk register. Even though the change itself sounded exciting, it also introduced an entirely new set of uncertainties that we needed to understand and control.

The biggest shift came from the engineering side. Underground stations involve tunneling, excavation, structural safety, and interactions with utilities and airport infrastructure. Because of this, we identified a new category of technical and construction risks for example, uncertainties around soil conditions, ventilation systems, and how the underground design would integrate with the airport's existing layout. These risks had not been part of our original assumptions, so we elevated their priority.

We also recognized coordination risks. An underground station brings additional stakeholders to the table, such as airport engineering teams, facility planners, and safety specialists. The more people involved, the greater the chance of misalignment. To mitigate this, we planned a focused meeting specifically aimed at bringing the right experts together early so misunderstandings wouldn't snowball later.

Another area we monitored was data and modeling accuracy. Switching from an above-ground to an underground design meant revising parts of our engineering and station databases. Any time you modify a core dataset, there's a risk of errors. To prevent this, we added an extra layer of data validation and peer review before using the updated information in our analysis.

Finally, we kept a close eye on schedule risk. Even though the change didn't immediately push us past any milestones, it had the potential to tighten our slack if not monitored. By updating dependencies in the schedule and checking task impacts regularly, we confirmed the additional work could be absorbed without affecting our critical path.

Overall, through continuous monitoring, early mitigation, and clear communication, we kept the new risks fully under control. The underground-station request did increase project complexity, but because we responded proactively, none of these risks escalated into problems that could delay or destabilize the project.

4. Resource & Cost Impact Analysis (EVM Summary)

After we integrated the underground station into our project, we stepped back and used Earned Value Management concepts to understand whether the change had meaningfully affected our resources or our overall performance. Even though the professor isn't looking for calculations, the logic behind EVM helps show that we approached the change in a disciplined way.

First, we looked at what we planned to have completed by this point in the project versus what we had actually completed. This allowed us to see if the underground station work had pulled us ahead, pushed us behind, or left us on track. We also reflected on the amount of effort the team had already spent, which is the cost side of EVM.

What we found was reassuring: the additional underground design work created only a small bump in effort, and because the change was caught early and coordinated well, it didn't disrupt any major milestones. Our actual progress stayed aligned with what we expected, and our resource usage stayed within a comfortable range. In other words, the change expanded the scope slightly, but the team was able to absorb that extra work smoothly.

The key takeaway from our EVM check was that the project remained under control. We didn't need to reset or re-baseline the project, because our schedule performance and our resource performance stayed within acceptable limits. The structured way in which we managed the change, documenting it, approving it, and adjusting our plan, allowed us to keep both cost and schedule impacts minimal.

Overall, the EVM assessment confirmed that even with the underground-station addition, the project continued moving forward on track, with a good balance between planned work, completed work, and the resources used along the way.

Cost Change

Assumptions (same policy as baseline)

- Blended labor rate **$85/hr**
- **Indirects = 30%** of direct labor
- **Contingency** = **15%** of (Direct+Indirect) for outreach/docs/PM; **20%** for station data & analysis (more uncertainty)
- Workshop logistics **$300** (room/refreshments/printing). Indirects applied to labor only.

Incremental cost by affected deliverable (Control Account)

Deliverable	Work Added (summary)	Direct $	Indirect (30%)	Cont.	Other	Total
D3 Scoping & Outreach	Plan/facilitate 1 DIA workshop + synthesis (18h)	1,530	459	298 (15%)	$300 exp	2,587
D4 Methodology & Existing Conditions *(4.5 Station DB)*	Add DIA station to property/station DB; access/land use checks (26h)	2,210	663	575 (20%)	–	3,448
D5 Preliminary Service Scenarios	Add DIA terminus option to scenarios (18h)	1,530	459	298 (15%)	–	2,287
D6 Alternatives Analysis	Ridership/ops/capex deltas for DIA option; evaluation updates (40h)	3,400	1,020	884 (20%)	–	5,304
D7 Feasibility Determination	Pro-forma & benefits deltas (16h)	1,360	408	265 (15%)	–	2,033
D8 Documentation	Update Impl/Business Plan	1,360	408	177	–	1,945

& Deliverables	sections (16h)			(10%)		
PM Coordination (X)	Approvals/comms/close CR (4h)	340	102	66 (15%)	–	508
						18,113

5. Expected Results & Final Recommendations

Introducing the underground station requirement ultimately strengthened our project and made our analysis more realistic. By adapting our work to reflect this new expectation, we positioned the study to better align with long-term transportation planning at the airport. Stakeholders now have a clearer understanding of how an underground design could function, what constraints it might involve, and how it would integrate with the broader rail system. This makes our final deliverables more relevant and more useful for decision-makers.

From an internal standpoint, the way we handled the change reinforced the value of having a structured project-management approach. Even though the request came unexpectedly, our team was able to respond quickly, adjust the necessary work packages, and still protect our schedule and resources. The change also improved collaboration bringing in additional engineering perspectives and ensuring stakeholder voices shaped the final concept.

Looking ahead, our main recommendations are to continue treating scope changes formally, even when they seem small. This type of discipline keeps the team aligned and prevents issues from growing. We also recommend maintaining early and proactive engagement with stakeholders, because changes often come from insights they raise during the process. Finally, building a small buffer into schedule and resources can make it much easier to absorb design adjustments, which are common in feasibility studies.

Overall, the project remains on track, within its planned scope and timeline, and we are confident moving into the final reporting phase. The underground station change did not derail the project; instead, it enriched the quality of our results and gave us a stronger, more forward-looking set of alternatives to present.

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